

Layout Design, Fabrication and Characterization of n-Channel MOSFET

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Abstract

Primary force behind evolution of semiconductor industry is MOSFET, after its first IC based experimental demonstration in 1960s. It is fundamental working unit of most of the electronic circuits like cells in living being. Number of devices on a single chip has been increased with rapid phase with evolution of new technology of materials. Today there are more than 4 billion transistors are there on a single chip with effective channel length of 45 nm and beyond. A lot complexity has been involved in device fabrication. This paper deals with detailed design of layout, fabrication and characterization of n-channel metal gate MOSFET fabricated on <100>, p-type silicon substrate using standard process steps and minimum mask levels.

Keywords: Depletion, e-beam lithography, mask, drain characteristics, transfer characteristics

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INTRODUCTION

Lilienfield in 1930, first described how modern MOSFET work [1]. First experimental demonstration of MOSFET was done by Kahng and Atalla [2, 3] in 1960s using silicon as substrate material. This MOS technology was populated only after development of fabrication process of reasonable quality silicon dioxide by Snow, Grove, Deal and Sah in 1964 [4]. After this MOS based fabrication industry has never looked back and its growth is very rapid.

The minimum feature size in production has reached beyond 45 nm. With the advancement in fabrication technology and the scalability limits of the device structure, the silicon based MOS ICs have continually delivered performance gain along with the cost reduction to various ICs [5]. The advancements in photo-lithographic and e-beam lithography [6], and well controlled doping by ion implantation have led to VLSI revolution [5].

Early MOSFETs used aluminum (Al) as gate material. In the late 1960s poly-silicon replaced Al as gate material for MOS transistor [7]. The next major milestone was LOCOS (Localized Oxidation of Silicon)

based isolation technique [8]. The commercial products based on n-channel MOSFET using LOCOS isolation were dominant in the mid-seventies. This n-channel MOSFET based technology provided reliable and scalable devices with reasonable manufacturing cost. Scaling of MOS devices has been started in 1970s and Dennard et al. in 1974 has formulated scaling rules [9, 10]. P-channel MOSFET that resulted in evolution of CMOS devices has also been started in same decade. The CMOS technology has revolutionized the state of art of the semiconductor IC industry due to its high noise immunity and almost nil static power dissipation. CMOS technology became the dominant technology for the VLSI chips 1980s [11].

BiCMOS technology has also been arisen to combine speed with low static power dissipation [12, 13]. Scaling limit has reached to further scale down SiO₂ and silicon nitrides, so replacement of these materials have been proposed. High-k dielectrics have replaced SiO₂ and silicon nitrides. A number of literatures have been published on high-K material as replacement of oxide material [14–22]. Fabrication of multi-mask devices is very complex, and it involves several intricate fabrication steps. This paper presents layout

design, fabrication and electrical characterization of n-channel metal oxide semiconductor field effect transistor in detail.

Layout and Mask Generation

First and foremost step towards fabrication of any microelectronic device is, design of layout. From this layout mask is fabricated. Physical layout is prepared according to particular technology and for the microelectronic circuit that need to be fabricated. These layouts are prepared according to a set of rules called layout design rules that take care of various geometric constraints [23]. These rules are used to achieve maximum reliability and yield by in given silicon area for that particular technology. There are two different types of layout design rules: micron based rules and lambda based rules [22].

Figure 1 shows basic design steps of layout formation. Layout editor (L-Edit) of Tanner Tools has been used for layout design with morban 20 technology file. Figure 2 shows complete layout of metal gate MOSFET with all mask layers used to fabricate mask for MOSFETs.

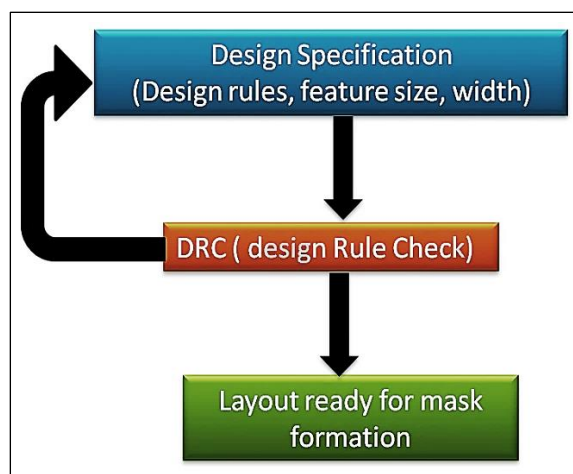


Fig. 1: Basic Design Steps of Layout Formation.

First Mask layer

Purpose of first mask layer is to open the oxide for source and drain region. Figure 3 shows first level mask. Window in SiO_2 layer for source and drain regions are open by photolithography using this mask. Buffered HF was used to etch out SiO_2 to create windows for source and drains.

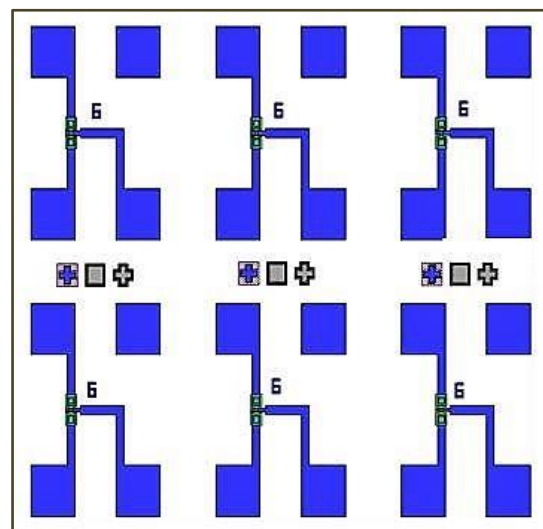


Fig. 2: Layout of Metal Gate MOS Transistor.

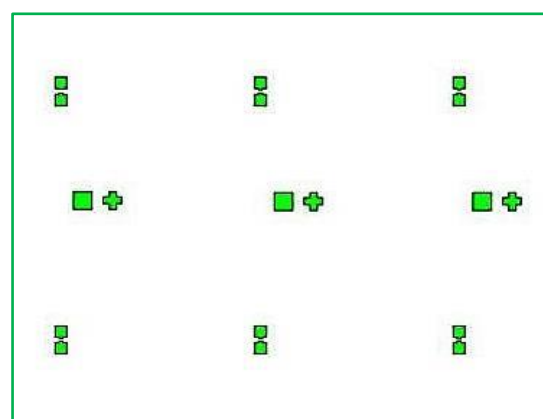


Fig. 3: First Mask Layer.

Second Mask Layer

Pattern for second level lithography mask is shown in Figure 4. The purpose of second level mask layer is to grow thin gate oxide. Buffered HF is used to remove the thick oxide between source and drain. High quality thin gate oxide is grown by dry oxidation to provide better control of gate over channel.

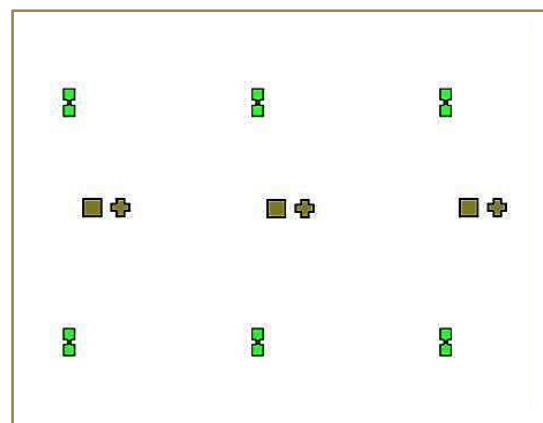


Fig. 4: Second Level Mask.

Third Level Mask

Third level mask of lithography is shown in Figure 5. The purpose of third masking layer is to open the source and drain window.



Fig. 5: Third Level Mask of Lithography.

Fourth Masking Layer

The purpose of fourth mask layer is to take the contact from drain, gate and source regions. Lithography mask pattern for fourth level is shown in Figure 6.

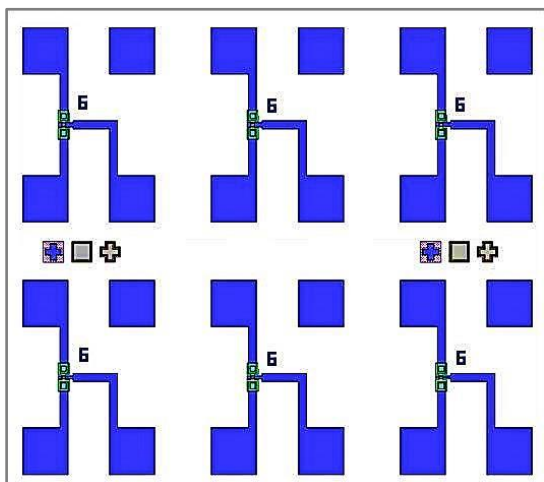


Fig. 6: Lithography mask pattern for fourth level.

Mask Formation

After completion of layout design each mask layer is transferred to a laser pattern writer and mask was transferred to glass mask.

Device Fabrication

Devices were fabricated on <100> p-type single side polished 4-inch diameter silicon wafers having resistivity window of 1–10 Ohm-cm. Wafers were cleaned using RCA cleaning. Wafers were dipped in HF to remove

native oxide formed during RCA cleaning before loading them in oxidation furnace. To grow thick field oxide of 5000 Å, wafer was oxidized in oxidation furnace at 1050°C. Dry-wet-dry cycle was followed to get good quality oxide. Oxygen flow rate for oxidation was 900 sccm. In dry-wet dry cycle, timing was 30 min -60 min -30 min respectively. For all the photolithography steps, positive photoresist, MICROPOSIT S1813 G2 and developer MICROPOSIT MF-319 were used. MBJ4 of SUSS MicroTec was used for photolithography. Basic process steps followed are shown in Figure 7. Aluminum metallization was done by sputtering process using 99.99% pure aluminum target.

Source and drain diffusion junctions were formed by diffusion of phosphorus, which is an n-type dopant. Solid state diffusion was done in two steps: pre-deposition and drive in. In pre-deposition process, maximum concentration of impurities diffused is at the wafer surface. To place the dopant atom at desired depth, drive in process is used. Diffusion process is started from pre-deposition step. Drive in was done by annealing of wafer at higher temperature (1050°C) in the nitrogen ambient. To make contacts to drain, gate and source, 300 nm thick aluminum was deposited on whole wafer by DC magnetron sputtering in argon plasma of 19.8 sccm. Substrate temperature of 35°C, sputtering pressure of 0.02 mbar and 120 W DC power was used for sputtering. After this, photolithography was performed to pattern the electrodes.

RESULTS

Keithley SCS 4200 semiconductor-characterization system was used for electrical characterization of fabricated n-channel MOSFET devices. Figure 8 shows drain characteristics of n-channel MOSFET. Drain current of the MOSFET was measured by varying drain voltage from 0 V to 12 V for gate to source voltage of 1 V, 2 V, 3 V, 4 V, 5 V and 6 V. Drain current increases with increasing drain to source and gate to source voltage. Threshold voltage of the device was found to be -1.65 V. Theoretical value of threshold voltage is -1.73 volts. These values are different from each other by 0.8 volts.

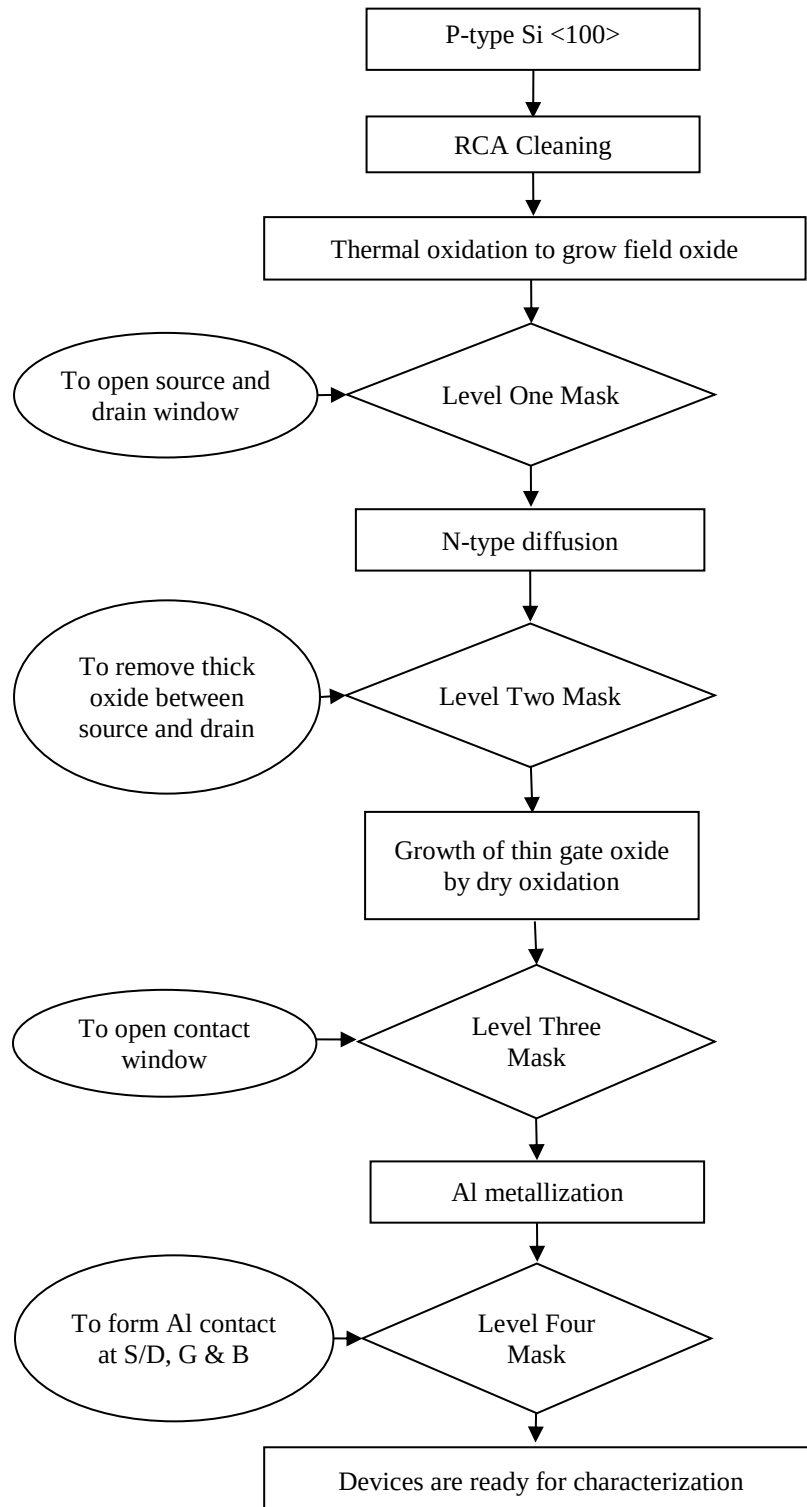


Fig. 7: Basic Steps of Process Flow for Fabrication of MOSFET.

This difference is there due to presence interface charges at the silicon and oxide interface. Due to presence of these charges, flat-band voltage arises that makes the difference between the two values. It can be observed from the figure that after drain saturation voltage, drain current is not constant

but increases with different slope. It means that there is enough channel length modulation that causes this finite slope in beyond saturation region. Sub-threshold slope of the device was found to be 66.85 mV/decade very close to 70 mV/decade. Trans-conductance of the device was measured as 0.27 mA/V.

Transfer characteristics of n-channel MOSFET were also measured. Figure 9 shows transfer characteristic curve of one of the fabricated devices. Drain current was measured by varying gate to source voltage from -2 to 8 V for two different drain to source voltages of 4 V and 8 V.

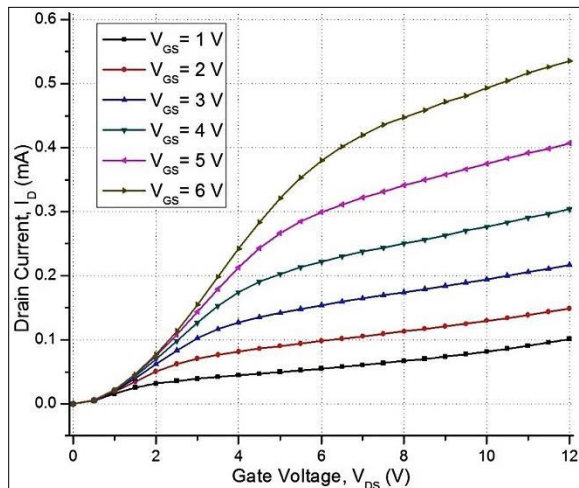


Fig. 8: Drain Characteristics of n-channel MOSFET.

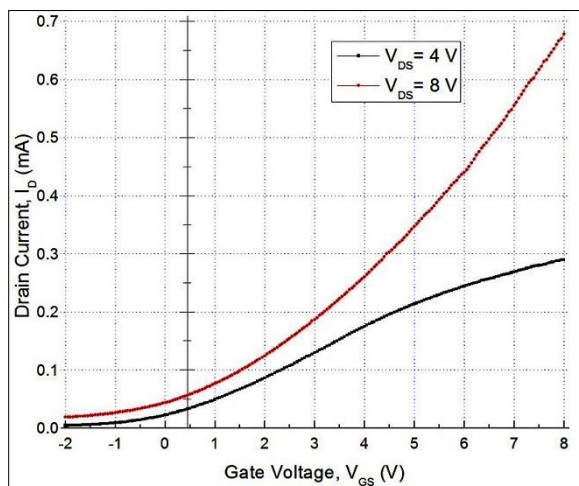


Fig. 9: Transfer Characteristics of n-channel MOSFET.

CONCLUSION

N-channel depletion type metal gate MOSFET has been fabricated successfully and characterized also. Drain characteristic and transfer characteristic were measured. Extracted threshold voltage of fabricated device is -1.65 V from the measured characteristics. Negative threshold voltage is reported for the devices fabricated with standard fabrication steps. Special

implantation is required to fabricate enhancement type devices.

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